MAT31-2

Probability & Statistics II

Continuous variables, limit theorems, and hypothesis testing

From counting outcomes to integrating densities — and from describing chance to testing claims with it.

Albert School — 30 hours

Contents

Preface

In **Probability & Statistics I** (MAT21-2) you learned to reason about uncertainty when outcomes can be **listed**: a die shows one of six faces, a box holds a whole number of defects, ten flips produce between zero and ten heads. You built probability mass functions, computed expectations as weighted sums $\mathbb{E}[X] = \sum_x x\, p_X(x)$, measured spread with the variance and the König–Huygens formula $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$, met the Bernoulli, binomial, geometric and Poisson families, and took your first steps in inference with maximum likelihood and Bayesian updating.

But most quantities we measure are not counts. A waiting time, a height, a temperature, a measurement error — these vary **continuously**, taking any value in an interval. No list of outcomes can capture them, and no sum can compute their averages. The tool that replaces the sum is the **integral**, which you built in **Foundations** (MAT11-1): the Fundamental Theorem of Calculus, integration by parts and substitution, and the limit-of-Riemann-sums picture of area. This course is, in large part, the discrete theory of MAT21-2 re-told with $\int$ in place of $\sum$ and a **density** in place of a mass function. If a manipulation here feels unfamiliar, the gap is usually in the calculus of MAT11-1, not in the probability.

Three deeper ideas then make their entrance, and they are the reason continuous probability matters so much for data science. First, the **limit theorems** — the Law of Large Numbers and the Central Limit Theorem — explain **why averages are trustworthy** and **why the bell curve is everywhere**. Second, **Monte Carlo methods** turn those theorems into a computational engine: when a probability is too hard to integrate, simulate it. Third, the machinery of **hypothesis testing** and **confidence intervals** — finally within reach now that we have the normal distribution — gives a disciplined way to draw conclusions from data and to state honestly how uncertain those conclusions are.

Our method is unchanged from MAT21-2: every idea arrives as a **concrete example first** and the **general statement second**, we name the mistakes students reliably make, and we keep sight of the modelling question — **what real situation does this object describe?** — rather than treating formulas as things to memorise.

1 Generalized integrals

A continuous random variable will spread its probability over an interval, and total probability must equal 1. When that interval is the whole real line — as for the normal distribution — computing "total probability" means integrating over an **unbounded** region. Ordinary definite integrals $\int_a^b$ run between two finite numbers; we first need to make sense of integrals that run to infinity or over a function that blows up. These are **generalized** (or **improper**) integrals.

The OP asks for **intuition and standard reference cases**, not the full convergence machinery (that is deferred to the Advanced track in S3). So we proceed by example and by a single guiding idea: **an improper integral is a limit of ordinary ones.**

1.1 Integrating to infinity

> **Worked example — the area under a decaying tail.** Consider $f(x) = 1/x^2$ for $x \geq 1$. Its graph hugs the axis as x grows, but never touches it — the region under it stretches infinitely far to the right. Does it nonetheless enclose a **finite** area? Integrate up to a finite cutoff b and watch:
>
> $$\int_1^b \frac{1}{x^2}\,\mathrm{d}x = \left[-\frac{1}{x}\right]_1^b = 1 - \frac{1}{b}.$$
>
> As $b \to \infty$ the term $1/b$ vanishes, leaving area exactly 1. An infinitely long region can have finite area.

Definition 1 (improper integral on an unbounded interval) If $\int_a^b f(x)\,\mathrm{d}x$ exists for every $b > a$, we define

$$\int_a^\infty f(x)\,\mathrm{d}x = \lim_{b\to\infty}\int_a^b f(x)\,\mathrm{d}x.$$

When the limit exists and is finite, the integral **converges**; otherwise it **diverges**. Integrals to $-\infty$, and the doubly-infinite $\int_{-\infty}^{\infty} f = \int_{-\infty}^{c} f + \int_c^\infty f$, are defined the same way.

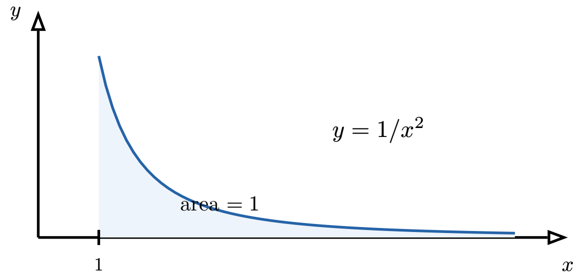

Figure 1: The improper integral $\int_1^\infty x^{-2}\,\mathrm{d}x$. The region is unbounded to the right yet has finite area 1, because the tail thins fast enough. We compute it as the limit of the finite areas $1 - 1/b$ as the cutoff $b \to \infty$.

The decisive question is **how fast the tail decays**. Compare $1/x^2$ with $1/x$:

$$\int_1^b \frac{1}{x}\,\mathrm{d}x = [\ln x]_1^b = \ln b \to \infty.$$

The area under $1/x$ is **infinite** — the curve decays too slowly. This is the canonical reference family:

Proposition 2 (the p-integral at infinity)

$$\int_1^\infty \frac{1}{x^p}\,\mathrm{d}x \quad \text{converges} \iff p > 1, \quad \text{and then equals } \frac{1}{p-1}.$$

Proof. For $p \neq 1$, $\int_1^b x^{-p}\,\mathrm{d}x = \left[x^{1-p}/(1-p)\right]_1^b = (b^{1-p}-1)/(1-p)$. If $p > 1$ the exponent $1-p$ is negative, so $b^{1-p} \to 0$ and the limit is $1/(p-1)$. If $p < 1$ the exponent is positive and $b^{1-p} \to \infty$. The borderline $p = 1$ gives $\ln b \to \infty$ as above. $\square$

A second indispensable case is the **exponential tail**, which decays faster than any power and will underlie the exponential and normal distributions:

$$\int_0^\infty e^{-\lambda x}\,\mathrm{d}x = \left[-\frac{1}{\lambda}e^{-\lambda x}\right]_0^\infty = \frac{1}{\lambda} \quad (\lambda > 0).$$

1.2 Integrating an unbounded function

The other way an integral turns improper is when the **integrand** blows up at an endpoint, even over a finite interval. The cure is identical — retreat to a cutoff and take a limit.

Worked example — a spike at the origin. The function $1/\sqrt{x}$ is unbounded as $x \to 0^+$. Yet

$$\int_0^1 \frac{1}{\sqrt{x}}\,\mathrm{d}x = \lim_{a\to 0^+}\int_a^1 x^{-1/2}\,\mathrm{d}x = \lim_{a\to 0^+}\left[2\sqrt{x}\right]_a^1 = \lim_{a\to 0^+}\left(2 - 2\sqrt{a}\right) = 2.$$

The spike is integrable: its area is finite. By contrast $\int_0^1 (1/x)\,\mathrm{d}x = \lim_{a\to 0^+}(-\ln a) = +\infty$ diverges.

Notice the mirror image of the tail case: near 0, $1/x^p$ converges exactly when $p < 1$ — the opposite inequality. The intuition: at infinity you need **fast decay** (p large); at a spike you need a **mild singularity** (p small).

Common pitfall Do not evaluate an improper integral by blindly plugging in the limits as if they were ordinary numbers. Writing "$[-1/x]_1^\infty = 0 - (-1)$" gives the right answer here only because the limit happens to exist; with $\int_1^\infty (1/x)\,\mathrm{d}x$ the same reflex hides a divergence. **Always** set up the limit explicitly when an endpoint is ∞ or a point where f is undefined.

Remark A useful shortcut when you only need to know **whether** an integral converges, not its value, is **comparison**: if $0 \le f(x) \le g(x)$ and $\int g$ converges, so does $\int f$. For instance $\int_1^\infty e^{-x^2}\,\mathrm{d}x$ converges because $e^{-x^2} \le e^{-x}$ for $x \ge 1$ and the exponential integral converges — even though e^{-x^2} has no elementary antiderivative. This is exactly why the normal distribution's normalising constant has to be found by a trick rather than by direct integration.

Exercises

1. Decide convergence and find the value when finite: $\int_1^\infty x^{-3}\,\mathrm{d}x$, $\int_1^\infty x^{-1/2}\,\mathrm{d}x$, $\int_0^\infty e^{-2x}\,\mathrm{d}x$.
2. Show $\int_0^1 x^{-p}\,\mathrm{d}x$ converges iff $p < 1$, and find its value.
3. Use comparison to show $\int_1^\infty (\sin^2 x)/x^2\,\mathrm{d}x$ converges.
4. For which constant c is $f(x) = c\,e^{-3x}$ on $[0,\infty)$ a candidate for a probability density (i.e. $\int_0^\infty f = 1$)?

2 Continuous random variables

A discrete variable places lumps of probability on isolated points. A **continuous** variable smears probability over an interval, so that any **single** value has probability zero and only **ranges** carry weight. The lump-heights $p_X(x)$ are replaced by a **density** $f_X(x)$, and every sum $\sum_x$ becomes an integral $\int \mathrm{d}x$. This one substitution carries almost all of MAT21-2 across.

2.1 Density and cumulative distribution function

Worked example — a random point on a stick. Snap a 1-metre stick at a uniformly random point and let X be the distance of the break from the left end. Then X lands anywhere in $[0,1]$ with no preference. What is $P(X = 0.5)$ — exactly the midpoint? Zero: there are infinitely many points and no single one can carry positive probability. But $P(0.4 \le X \le 0.6) = 0.2$, the **length** of the sub-interval. Probability has become **length**, and more generally **area under a curve**.

Definition 3 (probability density function) A **continuous random variable** X is described by a **probability density function** (PDF) $f_X(x) \ge 0$ such that probabilities are **areas**:

$$P(a \le X \le b) = \int_a^b f_X(x)\,\mathrm{d}x,$$

and the total area is one, $\int_{-\infty}^{\infty} f_X(x)\,\mathrm{d}x = 1$.

The density is **not** a probability — it can exceed 1 (a sharp spike over a short interval). Only its **integral** over a range is a probability. The correct infinitesimal reading is $P(x \le X \le x + \mathrm{d}x) \approx f_X(x)\,\mathrm{d}x$: the density times a tiny width gives the probability in that sliver.

Common pitfall Because single points have probability zero, the endpoints never matter for a continuous variable: $P(a \le X \le b) = P(a < X < b)$. Strict and non-strict inequalities give the same answer — a relief after the fussy off-by-one bookkeeping of discrete sums. (This fails completely for discrete variables, where $P(X \le 3)$ and $P(X < 3)$ can differ by $p_X(3)$.)

As in the discrete case, the **cumulative distribution function** accumulates probability from the left — but now by integrating, not summing.

Definition 4 (cumulative distribution function) The **CDF** of X is $F_X(x) = P(X \le x) = \int_{-\infty}^{x} f_X(t)\,\mathrm{d}t$. It is non-decreasing, runs from $F_X(-\infty) = 0$ to $F_X(+\infty) = 1$, and — by the Fundamental Theorem of Calculus — recovers the density by differentiation:

$$F_{X'}(x) = f_X(x).$$

This last identity is the through-line of the whole chapter: **density and CDF are a derivative–integral pair**, exactly the FTC relationship of MAT11-1. To get probabilities from a density, integrate; to get the density from a CDF, differentiate; and $P(a \le X \le b) = F_X(b) - F_X(a)$.

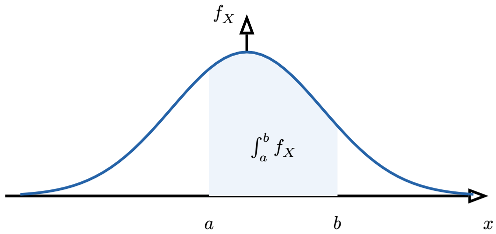

Figure 2: Probability as area. The shaded region under the density f_X between a and b equals $P(a \le X \le b)$. The whole region under the curve has area 1. Individual points, being slivers of zero width, carry zero probability.

2.2 Expectation, variance, and moments

Every definition from MAT21-2 transfers by turning $\sum_x x\,p_X(x)$ into $\int x\,f_X(x)\,\mathrm{d}x$. The **interpretation is identical** — expectation is the balance point of the density, variance its spread — only the computation changes from summing to integrating.

Definition 5 (expectation and variance) For a continuous variable X with density f_X,

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x\,f_X(x)\,\mathrm{d}x, \quad \mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2] = \mathbb{E}[X^2] - \mu^2,$$

where $\mu = \mathbb{E}[X]$, provided the integrals converge absolutely. The k-th **moment** is $\mathbb{E}[X^k] = \int x^k f_X(x)\,\mathrm{d}x$.

The **transfer theorem** (the "law of the unconscious statistician") also survives verbatim, and it is what lets us compute $\mathbb{E}[X^2]$ for the variance without first finding the distribution of X^2.

Theorem 6 (Transfer theorem) For any (measurable) function g,

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x)\, f_X(x)\,\mathrm{d}x.$$

Linearity of expectation, $\mathrm{Var}(aX+b) = a^2\,\mathrm{Var}(X)$, the König–Huygens formula, and variance-additivity for independent variables all hold word-for-word as in MAT21-2 — their proofs used only linearity of the sum, which is shared by the integral.

Worked example — mean and variance of a point on the stick. For the uniform break point, the density is $f_X(x) = 1$ on $[0,1]$ and 0 elsewhere (it must integrate to 1 over a unit interval). Then

$$\mathbb{E}[X] = \int_0^1 x \cdot 1\,\mathrm{d}x = \left[\frac{x^2}{2}\right]_0^1 = \frac{1}{2}, \quad \mathbb{E}[X^2] = \int_0^1 x^2\,\mathrm{d}x = \frac{1}{3},$$

so $\mathrm{Var}(X) = 1/3 - (1/2)^2 = 1/12$. The mean sits at the centre, as symmetry demands, and the standard deviation $\sigma = 1/\sqrt{12} \approx 0.29$ measures the typical distance from it.

Common pitfall A continuous variable can have **no** mean. If the tails of f_X are too heavy the defining integral diverges — the Cauchy density $f(x) = 1/(\pi(1+x^2))$ is the classic example, with $\int x f(x)\,\mathrm{d}x$ divergent. "Expectation exists" is a genuine condition, not a formality, exactly as absolute convergence was for discrete sums.

Exercises

5. Let $f(x) = cx^2$ on $[0,2]$, zero elsewhere. Find c, then $\mathbb{E}[X]$ and $\mathrm{Var}(X)$.
6. X has density $f(x) = 2x$ on $[0,1]$. Find its CDF F_X, then $P(X \leq 1/2)$ and the median (the value m with $F_X(m) = 1/2$).
7. Verify the transfer theorem numerically for $g(x) = x^2$ and $f(x) = 1$ on $[0,1]$ by also computing the density of $Y = X^2$ and integrating $y\, f_Y(y)$.

3 Classic continuous distributions

Just as four discrete families covered most discrete phenomena, a small catalogue of continuous distributions covers most continuous ones. We derive each from its **modelling assumption**, exactly the discipline of MAT21-2: the formula should feel **inevitable**, not memorised.

3.1 The uniform distribution

Definition 7 (uniform distribution) $X \sim \mathrm{Unif}(a, b)$ is "no preference within $[a, b]$": its density is constant,

$$f_X(x) = \frac{1}{b-a} \quad \text{for } a \le x \le b, \quad 0 \text{ otherwise} .$$

Then $\mathbb{E}[X] = (a + b)/2$ (the midpoint) and $\mathrm{Var}(X) = (b - a)^2/12$.

The constant height $1/(b - a)$ is forced by total area 1. The uniform is the continuous analogue of equiprobability, and — crucially for the last chapter — it is the output of a standard random-number generator, the raw material from which every other distribution will be simulated.

3.2 The exponential distribution and memorylessness

Worked example — waiting for the next bus. Buses arrive "at random" at an average rate of λ per hour, with no schedule. Let T be the time you wait. The Poisson process of MAT21-2 says the number of arrivals in time t is $\mathrm{Poi}(\lambda t)$, so the probability of **no** bus in $[0, t]$ is $e^{-\lambda t}$. Hence $P(T > t) = e^{-\lambda t}$, and the CDF is $F_T(t) = 1 - e^{-\lambda t}$. Differentiating gives the density.

Definition 8 (exponential distribution) $T \sim \mathrm{Exp}(\lambda)$ models the waiting time for a Poisson event of rate $\lambda > 0$:

$$f_T(t) = \lambda e^{-\lambda t}, \quad F_T(t) = 1 - e^{-\lambda t} \quad (t \ge 0).$$

Its mean and standard deviation are both $1/\lambda$: $\mathbb{E}[T] = 1/\lambda$ and $\mathrm{Var}(T) = 1/\lambda^2$.

The mean $1/\lambda$ is intuitive: at rate λ events per hour you wait on average $1/\lambda$ hours — the continuous echo of the geometric mean $1/p$. Indeed the exponential is the **continuous limit of the geometric**, and it inherits the geometric's defining quirk.

Theorem 9 (Memorylessness) The exponential is the only continuous distribution on $[0, \infty)$ that is **memoryless**:

$$P(T > s + t \mid T > s) = P(T > t) \quad \text{for all } s, t \ge 0.$$

Proof. $P(T > s + t \mid T > s) = P(T > s + t)/P(T > s) = e^{-\lambda(s+t)}/e^{-\lambda s} = e^{-\lambda t} = P(T > t)$. The exponential function turns a sum in the exponent into a product, which is exactly what conditioning needs. □

Common pitfall Memorylessness is deeply counter-intuitive and a notorious trap. A lightbulb with exponential lifetime that has already burned for 1000 hours is, **probabilistically**, as good as new: its remaining life has the same distribution as a fresh bulb's. This is a **property of the model**, appropriate for things that fail by sudden random shocks (radioactive decay, memoryless queues) but wrong for things that **wear out** (human lifespans, mechanical fatigue), whose hazard rises with age. Choosing the exponential asserts "no ageing".

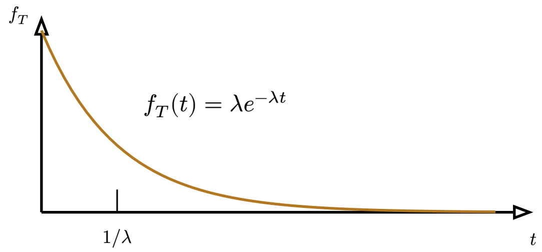

Figure 3: The exponential density for $\lambda = 1$. It is highest at $t = 0$ and decays at a constant relative rate — the geometric signature of "no ageing". The mean $1/\lambda$ lies to the right of the peak because the long right tail pulls the balance point outward.

3.3 The normal distribution

The normal (Gaussian) distribution is the centrepiece of the course. It will emerge **inevitably** from the Central Limit Theorem two chapters from now; here we meet its shape and learn to compute with it.

Definition 10 (normal distribution) $X \sim \mathcal{N}(\mu, \sigma^2)$ has the bell-shaped density

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$$

Its parameters are exactly its mean and variance: $\mathbb{E}[X] = \mu$, $\mathrm{Var}(X) = \sigma^2$. The curve is symmetric about μ, with inflection points at $\mu \pm \sigma$.

The two parameters do the obvious geometric jobs: μ **locates** the peak, σ sets the **width**. The prefactor $1/\big(\sigma\sqrt{2\pi}\big)$ is the normalising constant that makes the total area 1 — and as the remark in Chapter 1 hinted, the integral $\int e^{-x^2/2}\,\mathrm{d}x = \sqrt{2\pi}$ cannot be found by an elementary antiderivative; it requires a special trick (squaring and switching to polar coordinates).

3.3.1 Standardization

Because no antiderivative exists, normal probabilities are read from a **table** (or a calculator). We cannot tabulate every (μ, σ), so we reduce **every** normal to a single one by **standardizing**.

Theorem 11 (Standardization) If $X \sim \mathcal{N}(\mu, \sigma^2)$, then

$$Z = \frac{X-\mu}{\sigma} \sim \mathcal{N}(0, 1),$$

the **standard normal**. Consequently

$$P(X \le x) = P\left(Z \le \frac{x-\mu}{\sigma}\right) = \Phi\left(\frac{x-\mu}{\sigma}\right),$$

where Φ is the standard normal CDF, the tabulated function.

Subtracting μ re-centres the variable at 0; dividing by σ rescales it to unit spread. The transformed value $z = (x-\mu)/\sigma$ — the ***z*-score** — answers "how many standard deviations above the mean is x?", a unit-free question whose answer is the same for every normal.

> **Worked example — test scores.** Exam marks are $\mathcal{N}(\mu = 500, \sigma^2 = 100^2)$. What fraction of candidates score above 650? Standardize: $z = (650-500)/100 = 1.5$. So $P(X > 650) =$

> $P(Z > 1.5) = 1 - \Phi(1.5) \approx 1 - 0.9332 = 0.0668$ — about 6.7%. The raw mark 650 became the universal statement "1.5 standard deviations above average", and the table did the rest.

A compact table of $\Phi(z) = P(Z \le z)$ for the standard normal:

z	0.0	0.5	1.0	1.5	2.0	2.5	3.0
$\Phi(z)$	.5000	.6915	.8413	.9332	.9772	.9938	.9987

For **negative** z, symmetry gives $\Phi(-z) = 1 - \Phi(z)$ — the table need only list $z \ge 0$. Two values are worth memorising because they recur in every confidence interval: $\Phi(1.645) = 0.95$ and $\Phi(1.96) = 0.975$.

Common pitfall The table gives $\Phi(z) = P(Z \le z)$, the area to the **left**. For a **right** tail use $P(Z > z) = 1 - \Phi(z)$; for a **two-sided** range use $P(-z \le Z \le z) = 2\Phi(z) - 1$; and for **negative** arguments use $\Phi(-z) = 1 - \Phi(z)$. Forgetting which tail you are in is the single most common error in this chapter — always sketch the bell curve and shade the region you want before reaching for a number.

3.3.2 The 68–95–99.7 rule

Proposition 12 (the empirical rule) For any normal variable,

$$P(\mu - \sigma \le X \le \mu + \sigma) \approx 68\%, \quad P(\mu - 2\sigma \le X \le \mu + 2\sigma) \approx 95\%, \quad P(\mu - 3\sigma \le X \le \mu + 3\sigma) \approx 99.7\%.$$

These follow directly from the table: $2\Phi(1) - 1 \approx 0.68$, $2\Phi(2) - 1 \approx 0.954$, $2\Phi(3) - 1 \approx 0.997$. The rule is the fastest sanity-check in statistics: a normal value almost never strays beyond three standard deviations, and a value two deviations out is already a one-in-twenty event.

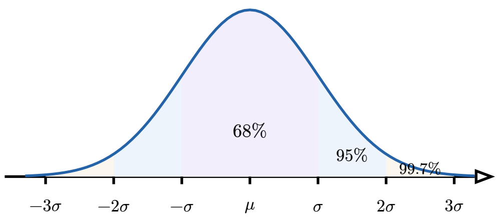

Figure 4: The 68–95–99.7 rule. The three nested shaded bands hold about 68%, 95% and 99.7% of a normal distribution's mass within one, two and three standard deviations of the mean. A value beyond 3σ is genuinely rare.

3.4 Two distributions for inference: chi-squared and Student-t

The last two families exist to **support statistical inference** — they describe how sample statistics behave — so we introduce them briefly now and use them later.

Definition 13 (chi-squared distribution) If $Z_1, \ldots, Z_k$ are independent standard normals, then

$$\chi_k^2 = Z_1^2 + \cdots + Z_k^2$$

has the **chi-squared distribution with k degrees of freedom**. It lives on $[0, \infty)$, has mean k and variance $2k$, and is right-skewed.

Being a sum of squares, χ^2 is the natural distribution of **sample variances** and of "goodness-of-fit" discrepancies. As k grows it becomes more symmetric — a foretaste of the Central Limit Theorem acting on the squared normals.

Definition 14 (Student-t distribution) If $Z \sim \mathcal{N}(0, 1)$ and χ^2_k are independent, then

$$T = \frac{Z}{\sqrt{\chi^2_k / k}}$$

has the **Student-t distribution with k degrees of freedom**. It is symmetric and bell-shaped like the normal but with **heavier tails**; as $k \to \infty$ it converges to $\mathcal{N}(0, 1)$.

The t-distribution is what we must use in place of the normal when the population variance is **unknown** and estimated from a **small** sample: the extra uncertainty in estimating σ fattens the tails, so t produces slightly wider — more honest — confidence intervals. For large samples the difference vanishes and the normal returns.

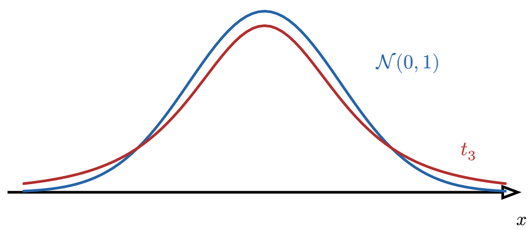

Figure 5: Standard normal (blue) versus Student-t with 3 degrees of freedom (red). Both are symmetric and bell-shaped, but t has a lower peak and **heavier tails**: extreme values are more probable, reflecting the extra uncertainty of a small sample. As the degrees of freedom grow, t tightens onto the normal.

Exercises

8. $X \sim \text{Unif}(2, 8)$. Find $\mathbb{E}[X]$, $\text{Var}(X)$, and $P(3 \le X \le 5)$.

9. Call durations are $\text{Exp}(\lambda)$ with mean 4 minutes. Find λ, then $P(T > 6)$ and $P(T > 10 \mid T > 4)$ — and explain why the second equals $P(T > 6)$.

10. Heights are $\mathcal{N}(170, 8^2)$ cm. What fraction lie between 162 and 186 cm? Above 190 cm?

11. Using the empirical rule only, estimate the IQ range ($\mathcal{N}(100, 15^2)$) containing the middle 95% of people.

4 The Law of Large Numbers

We now reach the theorems that justify the entire practice of statistics. The first answers the most basic question one could ask of probability: **if I repeat an experiment many times, does the average outcome settle down to the expected value?**

4.1 Statement and interpretation

Worked example — the running average of coin flips. Flip a fair coin and record the **running proportion** of heads after $1, 2, 3, \ldots$ flips. Early on it lurches — 0, then 0.5, then 0.33 — but as flips accumulate it stops wandering and homes in on 0.5. Not because some force "corrects" imbalances, but because each new flip contributes an ever-smaller nudge to an ever-larger pile. The average is **self-stabilising**.

Theorem 15 (Law of Large Numbers) Let $X_1, X_2, \ldots$ be independent and identically distributed with mean μ and finite variance. Their **sample mean**

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$$

converges to μ as $n \to \infty$: for any tolerance $\varepsilon > 0$,

$$P\left(|\overline{X}_n - \mu| > \varepsilon\right) \to 0.$$

Proof. (sketch, via the indicator estimate of MAT21-2′s variance machinery) By linearity $\mathbb{E}[\overline{X}_n] = \mu$, and by variance-additivity for independent variables $\mathrm{Var}\left(\overline{X}_n\right) = \sigma^2/n \to 0$. So the sample mean is an unbiased estimator of μ whose spread collapses. Chebyshev's inequality $P\left(|\overline{X}_n - \mu| > \varepsilon\right) \le \mathrm{Var}\left(\overline{X}_n\right)/\varepsilon^2 = \sigma^2/(n\varepsilon^2)$ then drives the probability to zero. □

The proof reveals **why** it works and connects straight back to MAT21-2: the variance of an average of n independent copies is $1/n$ of a single one's, so spread shrinks like $1/n$ — the same $p(1-p)/n$ we found for the sample proportion. The LLN is the formal promise that **more data means a more reliable average**.

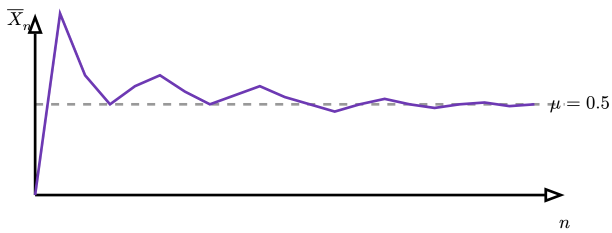

Figure 6: The Law of Large Numbers in action: the running proportion of heads swings wildly at first, then converges to $\mu = 0.5$. The amplitude of the swings shrinks like $1/\sqrt{n}$ — the convergence is real but gradual.

4.2 The Gambler's Fallacy

Common pitfall The LLN says the **average** converges; it does **not** say the universe keeps a ledger and repays deficits. After a run of 10 heads, the **Gambler's Fallacy** insists tails is now "due". It is not: the coin is memoryless, so the next flip is still 50/50. Convergence happens by **dilution**, not **compensation** — the early surplus of heads is not cancelled by future tails, it is simply **swamped** by the growing denominator n. An absolute imbalance can even **grow** while the proportion converges. Confusing "the proportion tends to $1/2$" with "the counts must even out" is the most expensive misreading of probability ever committed.

4.3 Implication for sampling

The LLN is the licence to **estimate by sampling**. The sample proportion of voters favouring a candidate converges to the true proportion; the sample mean income converges to the population mean. It guarantees that the estimate is **eventually** right — but it is silent on **how fast**, and on **how wrong** a finite sample might be. For that quantitative question we need the next theorem.

Exercises

12. A die is rolled n times. To what value does the average of the rolls converge? Justify with the LLN.
13. Explain to a casino patron, using the dilution idea, why ten reds on roulette do not make black more likely next spin.
14. The sample mean of n i.i.d. variables with variance $\sigma^2 = 4$ must have standard deviation below 0.1. How large must n be?

5 The Central Limit Theorem

The LLN tells us the sample mean lands on μ; the **Central Limit Theorem** tells us **how it is distributed around μ** on the way there. Its answer is astonishing and universal: **whatever the shape of the original distribution, the sample mean is approximately normal.** This is why the bell curve is everywhere.

5.1 Statement

> **Worked example — averaging dice.** A single die is uniform on $1, ..., 6$ — flat, nothing like a bell. Average two dice and the distribution becomes triangular, peaked at $7/2$. Average ten, and a smooth bell emerges. The flatness of the individual die is **forgotten**; only its mean and variance survive into the shape of the average. Summing many independent things **manufactures** normality.

Theorem 16 (Central Limit Theorem) Let $X_1, ..., X_n$ be i.i.d. with mean μ and finite variance σ^2. As $n \to \infty$, the standardized sample mean converges in distribution to a standard normal:

$$\frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}} \to \mathcal{N}(0, 1).$$

Equivalently, for large n, $\overline{X}_n \approx \mathcal{N}(\mu, \sigma^2/n)$ and the sum $S_n = \sum X_i \approx \mathcal{N}(n\mu, n\sigma^2)$.

The quantity $\sigma/\sqrt{n}$ — the standard deviation of the sample mean — is called the **standard error**. The $\sqrt{n}$ is the heartbeat of statistics: to halve the error you need **four times** the data. The LLN said the error $\to 0$; the CLT says it does so at rate $1/\sqrt{n}$ and with a **normal** shape, which is exactly what makes confidence intervals computable.

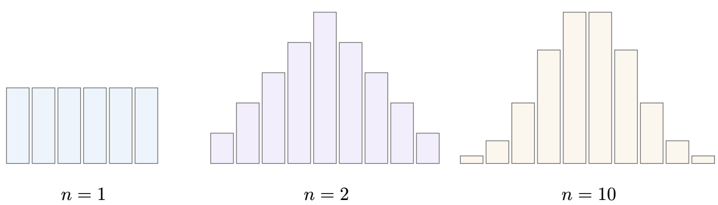

Figure 7: The Central Limit Theorem building a bell from a flat die. The distribution of the **sum** of n dice: flat for $n = 1$, triangular for $n = 2$, and visibly bell-shaped by $n = 10$. The original shape is irrelevant — only μ and σ^2 survive into the limiting normal.

5.2 Normal approximation of the binomial, with continuity correction

The first and most useful application: a binomial $X \sim \text{Bin}(n, p)$ is a sum of n i.i.d. Bernoulli(p) variables, so by the CLT it is approximately normal for large n:

$$\text{Bin}(n, p) \approx \mathcal{N}(np, np(1-p)).$$

This rescues us from summing dozens of binomial terms by hand. One subtlety: we are approximating a **discrete** variable (integer counts) by a **continuous** one, so we spread each integer k across the interval $[k-0.5, k+0.5]$ — the **continuity correction**.

> **Worked example — at least 60 heads in 100 flips.** Let $X \sim \text{Bin}(100, 0.5)$, so $np = 50$ and $\sqrt{np(1-p)} = \sqrt{25} = 5$. For $P(X \geq 60)$, the continuity correction replaces the cutoff 60 by 59.5:
>
> $$P(X \geq 60) \approx P\left(Z \geq \frac{59.5 - 50}{5}\right) = P(Z \geq 1.9) = 1 - \Phi(1.9) \approx 0.0287.$$
>
> Without the correction we would use 60, giving $z = 2.0$ and 0.0228 — a 20% relative error in the tail. The half-unit shift matters most precisely where we care most: in the tails.

Common pitfall The continuity correction direction depends on the inequality. For $P(X \geq k)$ use $k - 0.5$ (include the whole bar for k); for $P(X \leq k)$ use $k + 0.5$; for $P(X = k)$ use the band $[k - 0.5, k + 0.5]$. Dropping the correction is acceptable only when n is very large; for moderate n it is a real source of error. As a rule of thumb the approximation is trustworthy when $np \geq 5$ **and** $n(1-p) \geq 5$.

5.3 Choosing a sample size

Because the standard error is $\sigma/\sqrt{n}$, the CLT lets us **design** a study in advance: solve for the n that delivers a target precision. If we want the sample mean within ε of μ with 95% confidence, we need $1.96\,\sigma/\sqrt{n} \leq \varepsilon$, i.e.

$$n \geq \left(1.96\,\frac{\sigma}{\varepsilon}\right)^2.$$

This single inequality — precision improves with $\sqrt{n}$, so cost grows with the **square** of the precision demanded — governs the economics of every poll, survey and A/B test.

Exercises

15. A fair coin is flipped 400 times. Use the CLT (with continuity correction) to estimate $P(X \geq 220)$.

16. Component weights have mean 50 g and SD 4 g. A box holds 100 components. Approximate the probability the box weighs more than 5050 g.

17. How many voters must be polled so the sample proportion is within $\varepsilon = 0.03$ of the truth with 95% confidence? (Use the worst case $p(1-p) \leq 1/4$.)

6 Monte Carlo methods

The CLT does more than describe sampling — it **powers a computational method**. When a probability or an expectation is too hard to compute analytically (no closed-form integral, a tangled multidimensional region), we **simulate**: draw many random samples, average, and let the LLN and CLT do the rest. This is the **Monte Carlo** method, the workhorse of modern quantitative science.

6.1 Estimating an expectation by simulation

Worked example — estimating pi by throwing darts. Throw darts uniformly at the unit square $[0,1]^2$. A dart lands inside the quarter-circle $x^2 + y^2 \le 1$ with probability equal to that region's area, $\pi/4$. So if a fraction $\hat{p}$ of N darts land inside, then $4\hat{p}$ estimates π. With $N = 10000$ darts one typically gets $\pi \approx 3.14 \pm 0.02$ — no geometry, just counting.

Definition 17 (Monte Carlo estimator) To estimate $\theta = \mathbb{E}[g(X)]$, draw independent samples $X_1, \ldots, X_N$ from the distribution of X and average:

$$\hat{\theta}_N = \frac{1}{N} \sum_{i=1}^{N} g(X_i).$$

By the LLN $\hat{\theta}_N \to \theta$, and by the CLT its error is approximately normal with standard error $\sigma_g/\sqrt{N}$, where $\sigma_g^2 = \mathrm{Var}(g(X))$.

The estimator is unbiased ($\mathbb{E}\left[\hat{\theta}_N\right] = \theta$ by linearity) and its $1/\sqrt{N}$ error rate is the same $\sqrt{N}$ law as before: a Monte Carlo answer with one more digit of accuracy costs 100 times the computation. The method is general — it estimates **any** expectation, hence any probability (take g an indicator) and any integral (write it as an expectation).

6.2 The inverse transform method

To simulate, we must first **generate** samples from the target distribution. A computer supplies only $U \sim \mathrm{Unif}(0,1)$. How do we turn uniform noise into, say, an exponential variable? The key is the CDF.

Theorem 18 (Inverse transform method) Let F be a continuous, strictly increasing CDF. If $U \sim \mathrm{Unif}(0,1)$, then

$$X = F^{-1}(U)$$

has CDF F. Conversely, $F(X) \sim \mathrm{Unif}(0,1)$ for any continuous X.

Proof. $P(X \le x) = P(F^{-1}(U) \le x) = P(U \le F(x)) = F(x)$, using that F is increasing (so we may apply it to both sides) and that $P(U \le u) = u$ for a standard uniform. □

The picture: feeding a uniform height u on the vertical axis through the inverse CDF reads off the x whose accumulated probability is u. Regions where F rises steeply (high density) catch more of the uniform input, so they receive more samples — exactly as they should.

Worked example — simulating an exponential. The exponential CDF is $F(x) = 1 - e^{-\lambda x}$. Solve $u = 1 - e^{-\lambda x}$ for x: $x = -\ln(1-u)/\lambda$. So if $U \sim \mathrm{Unif}(0,1)$ then $X = -\ln(1-U)/\lambda$ is $\mathrm{Exp}(\lambda)$. (Since $1 - U$ is also uniform, $-\ln(U)/\lambda$ works just as well.) One logarithm turns a uniform into an exponential.

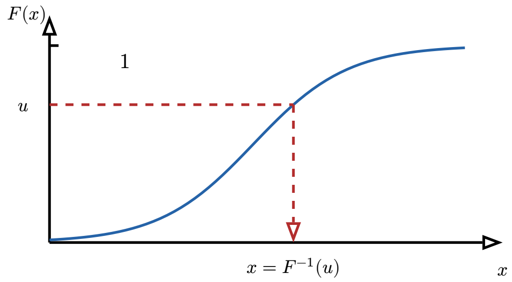

Figure 8: The inverse transform method. A uniform input $u \in (0, 1)$ on the vertical axis is mapped horizontally to the curve, then down to $x = F^{-1}(u)$. Steep parts of F (where the density is high) span more vertical room and so capture more uniform inputs — producing samples with the right density.

6.3 Confidence intervals for a Monte Carlo estimate

A simulation is worthless without an **error bar**. The CLT supplies one directly. Since $\hat{\theta}_N \approx \mathcal{N}(\theta, \sigma_g^2/N)$, and we estimate σ_g by the sample standard deviation s of the values $g(X_i)$, a 95% confidence interval for θ is

$$\hat{\theta}_N \pm 1.96 \cdot \frac{s}{\sqrt{N}}.$$

The half-width $1.96\, s/\sqrt{N}$ is the **standard error band**: it shrinks like $1/\sqrt{N}$, so reporting it tells the reader exactly how much to trust the simulation. **A Monte Carlo result without this band is incomplete.**

Exercises

18. Write the inverse-transform recipe to simulate $X \sim \mathrm{Unif}(a, b)$ from a standard uniform U.
19. You estimate $\mathbb{E}[g(X)]$ from $N = 1000$ samples and get mean 2.4 with sample SD $s = 1.5$. Give a 95% confidence interval. How large must N be to halve its width?
20. Describe a Monte Carlo scheme to estimate $\int_0^1 e^{-x^2}\, dx$ as an expectation, and state its standard error structure.

7 Hypothesis testing

We can now do **confirmatory** statistics: not merely estimate a quantity, but **test a claim** about it and quantify the evidence. Hypothesis testing is the disciplined procedure for deciding whether data are **surprising enough** under a default assumption to warrant rejecting it. The normal distribution, finally in hand, is what makes the arithmetic possible.

7.1 The logic: H_0, H_1, and a test statistic

Worked example — is the coin fair?. A coin is flipped 100 times and lands heads 62 times. Is it biased, or is 62 just luck? We adopt a **sceptical default** — "the coin is fair, $p = 0.5$" — and ask: **if that were true, how surprising is an outcome as extreme as 62?** Under fairness $X \sim \mathrm{Bin}(100, 0.5) \approx \mathcal{N}(50, 25)$, so the z-score is $(62 - 50)/5 = 2.4$. A result 2.4 standard errors out is rare — strong reason to doubt the default.

Definition 19 (null and alternative hypotheses) The **null hypothesis** H_0 is the default, no-effect claim we try to disprove (here $p = 0.5$). The **alternative hypothesis** H_1 is what we entertain if H_0 fails (here $p \neq 0.5$). A **test statistic** is a number computed from the data whose distribution **under H_0** is known; we reject H_0 when the statistic falls in a **rejection region** of values too extreme to attribute to chance.

The asymmetry is deliberate and important: we never **prove** H_0, we only **fail to reject** it. The burden of proof lies on the alternative, exactly as a court presumes innocence until evidence is strong enough to overturn it.

7.2 Significance level and the p-value

Definition 20 (significance level and rejection region) The **significance level** α (commonly 0.05) is the probability of rejecting H_0 **when it is actually true** — the false-alarm rate we are willing to tolerate. The **rejection region** is the set of test-statistic values whose total probability under H_0 equals α. For a two-sided z-test at $\alpha = 0.05$ it is $|z| > 1.96$.

Definition 21 (p-value) The ***p*-value** is the probability, **computed assuming H_0 is true**, of obtaining a result **at least as extreme** as the one observed. We reject H_0 exactly when $p \le \alpha$.

For the coin, the two-sided p-value is $P(|Z| \ge 2.4) = 2(1 - \Phi(2.4)) \approx 2(0.0082) = 0.0164$. Since $0.0164 < 0.05$, we reject fairness at the 5% level: an outcome this extreme would happen only about 1.6% of the time with a fair coin.

Common pitfall The p-value is the **probability of data this extreme given H_0** — written $P(\text{data} \mid H_0)$. It is **not** the probability that H_0 is true, $P(H_0 \mid \text{data})$. These are the two sides of Bayes' theorem from MAT21-2, and confusing them is the most damaging error in applied statistics. "$p = 0.02$" does **not** mean "there is a 2% chance the coin is fair"; it means "**if** the coin were fair, data this extreme would arise 2% of the time". A small p-value casts doubt on H_0; it does not measure H_0's probability.

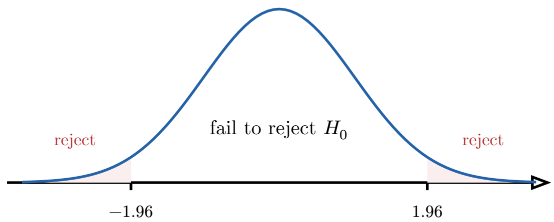

Figure 9: A two-sided z-test at $\alpha = 0.05$. Under H_0 the test statistic is standard normal; the two red tails beyond ± 1.96 together hold 5% of the mass. A statistic landing there is "too extreme to be chance", so H_0 is rejected. The p-value is the area beyond the **observed** statistic.

7.3 Type I and Type II errors, and power

Every test can err in two opposite ways, and the trade-off between them is the core design problem of testing.

Definition 22 (the two error types) A **Type I error** (false positive) is rejecting H_0 when it is true; its probability is the significance level α. A **Type II error** (false negative) is failing to reject H_0 when H_1 is true; its probability is denoted β. The **power** of the test is $1 - \beta$, the probability of **correctly** detecting a true effect.

	H_0 **true**	H_1 **true**
reject H_0	Type I error (α)	correct (power $1 - \beta$)
fail to reject	correct	Type II error (β)

The two errors pull against each other: lowering α (demanding stronger evidence to reject) shrinks false alarms but **raises** β, missing more real effects. The only way to reduce **both** at once is to gather more data — a larger n sharpens the sampling distributions and increases power at any fixed α. This is why power calculations decide sample sizes before an experiment is run.

Common pitfall "Fail to reject H_0" is **not** "accept H_0" or "prove no effect". A non-significant result may simply mean the test lacked the power to detect a real but small effect — **absence of evidence is not evidence of absence**. Reporting the power, or a confidence interval, is what distinguishes "we showed there is no effect" from "our study was too small to tell".

Exercises

21. A new drug is tested against placebo. State H_0, H_1, and describe a Type I and a Type II error in plain words, with their real-world costs.
22. A machine fills bottles to a target 500 ml ($\sigma = 5$). A sample of 25 has mean 497.5. Test $H_0 : \mu = 500$ at $\alpha = 0.05$ (two-sided): compute the z-statistic, the p-value, and conclude.
23. Explain, using the picture of two overlapping normal curves (under H_0 and under H_1), why increasing n raises power without raising α.

8 Confidence intervals

A hypothesis test answers a yes/no question; a **confidence interval** answers the more useful quantitative one: **what range of parameter values is plausible given the data?** Built from the same CLT, it carries its own precise — and famously misread — interpretation.

8.1 Construction from the CLT

Worked example — estimating average height. We measure $n = 100$ people and find sample mean $\overline{x} = 171$ cm, with known population SD $\sigma = 8$ cm. By the CLT $\overline{X} \approx \mathcal{N}(\mu, 8^2/100)$, so the standard error is $8/\sqrt{100} = 0.8$ cm. With 95% probability $\overline{X}$ lands within 1.96 standard errors of μ — and we can turn that around into a statement about μ.

Definition 23 (confidence interval for a mean) From a sample mean $\overline{x}$ with standard error $\sigma/\sqrt{n}$, a **95% confidence interval** for μ is

$$\overline{x} \pm 1.96 \cdot \frac{\sigma}{\sqrt{n}}.$$

The half-width $E = 1.96\,\sigma/\sqrt{n}$ is the **margin of error**. For a general confidence level $1-\alpha$, replace 1.96 by the critical value $z_{\alpha/2}$ with $\Phi\left(z_{\alpha/2}\right) = 1-\alpha/2$.

For the heights: $171 \pm 1.96(0.8) = 171 \pm 1.57$, i.e. $[169.4, 172.6]$ cm. When σ is unknown and estimated from a small sample, the t-distribution of Chapter 3 replaces the normal, widening the interval to account for the extra uncertainty.

8.2 What "95% confident" actually means

Common pitfall The interval $[169.4, 172.6]$ does **not** mean "there is a 95% probability that μ lies in it". Once computed, the interval either contains the fixed (but unknown) μ or it does not — there is no probability left. The 95% is a property of the **procedure**: if we repeated the whole sampling-and-computing process many times, about 95% of the intervals produced would capture μ. **Confidence is in the method, not in any single interval.** Saying "$P(\mu \in [169.4, 172.6]) = 0.95$" treats the fixed parameter as random — precisely the error this caveat exists to prevent.

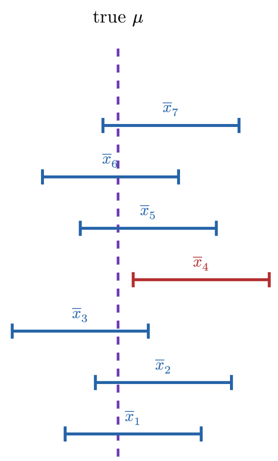

Figure 10: Seven samples, seven confidence intervals. Each brackets its own sample mean; the true μ (dashed line) is fixed. Most intervals capture μ, but one (red) misses entirely. Over many repetitions about 95% would capture it — that frequency **is** the confidence level. Any single interval simply does or does not.

8.3 Margin of error and sample size

The margin of error $E = z_{\alpha/2}\,\sigma/\sqrt{n}$ exposes the three levers of precision. A **narrower** interval demands a larger n (cost grows as E^{-2}), a smaller σ (less variable data), or a **lower** confidence level (accepting more misses). Inverting for n recovers the sample-size formula of the CLT chapter:

$$n \geq \left(z_{\alpha/2}\,\frac{\sigma}{E}\right)^2.$$

8.4 Duality with hypothesis tests

Confidence intervals and two-sided tests are **two views of one computation**.

Proposition 24 (CI–test duality) A two-sided test of $H_0 : \mu = \mu_0$ at level α rejects H_0 **if and only if** μ_0 lies **outside** the $(1-\alpha)$ confidence interval.

The interval is exactly the set of null values that the data would **not** reject — the "plausible" parameter values. This is often the better way to report a result: rather than a bare "reject / fail to reject", the interval shows **which** values remain tenable and **how precisely** the parameter has been pinned down. A test says whether μ_0 is plausible; the interval shows the whole range of plausible μ at once.

> **Worked example — reading the duality.** For the heights, the 95% interval is $[169.4, 172.6]$. A claim "$H_0 : \mu = 170$" would **not** be rejected at $\alpha = 0.05$, since 170 lies inside. A claim "$H_0 : \mu = 168$" **would** be rejected, since 168 lies outside. We never recomputed a test — the interval already encodes every two-sided test at once.

Exercises

24. A sample of $n = 64$ has mean 52 with known $\sigma = 12$. Construct a 95% and a 99% confidence interval for μ. Why is the 99% interval wider?

25. A poll of 1000 voters finds 52% support. Give the margin of error and the 95% interval for the true proportion. Is "50%" a plausible value?

26. How large a sample makes the margin of error for a mean at most 1, if $\sigma = 10$ and confidence is 95%?

27. State the test that is dual to a 90% confidence interval, and explain why a value just outside the interval would be rejected.

9 A first look at sequential decisions

We close with a glimpse of where probability meets **decision-making over time** — the mathematical core of reinforcement learning. The full algorithmic theory belongs to the S6 Math-for-ML capstone; here we build the single foundational identity, the **Bellman equation**, on one small, fully worked example. Everything rests on the expectation you have used throughout this course.

9.1 The Markov property

> **Worked example — a two-state machine.** A machine is each day either **Working** (W) or **Broken** (B). If working, it stays working tomorrow with probability 0.8 and breaks with probability 0.2. If broken, the operator must choose an **action**: **Repair** (cheap, returns it to working with probability 0.6) or **Replace** (costly, returns it to working for certain). The crucial structural fact: **tomorrow depends only on today's state and action — not on the full history of how the machine got here.**

Definition 25 (Markov property) A process has the **Markov property** if the distribution of the next state depends **only** on the current state (and action), not on the earlier history:

$$P(\text{next state} \mid \text{current state, all past}) = P(\text{next state} \mid \text{current state}).$$

The data of such a model are a **state space**, a **transition kernel** (the probabilities of moving between states under each action), a **reward** attached to states or transitions, and a **discount factor** $\gamma \in [0, 1)$.

The Markov property is the continuous-time cousin of the memorylessness you met for the exponential: the present **screens off** the past. It is what makes long-horizon planning tractable — you need only know where you are, not how you arrived.

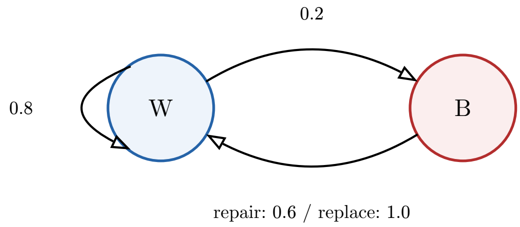

Figure 11: The two-state machine. From **Working** it breaks with probability 0.2; from **Broken** the chosen action (repair or replace) returns it to working with different probabilities. The next state depends only on the present state and action — the Markov property.

9.2 Value and the Bellman equation

How good is it to be in a given state? We measure it by the **value** — the total reward we expect to collect from now on, with future rewards **discounted** because a reward today is worth more than the same reward later.

Definition 26 (state value) The **value** $V(s)$ of a state s is the expected **discounted return** — the sum of present and future rewards, each future reward weighted by γ^t:

$$V(s) = \mathbb{E}[R_0 + \gamma R_1 + \gamma^2 R_2 + \cdots \mid \text{start in } s].$$

The discount $\gamma \in [0, 1)$ makes the infinite sum converge (a geometric series, as in MAT21-2) and encodes "sooner is better".

Now the key insight. The return splits into **today's reward** plus **everything from tomorrow on** — and "everything from tomorrow on" is just γ times the value of wherever we land. Taking the best action gives the **Bellman optimality equation**.

Theorem 27 (Bellman optimality equation) The optimal value satisfies, for each state s,

$$V(s) = \max_a \left[r(s, a) + \gamma \sum_{s'} P(s' \mid s, a)\, V(s') \right],$$

read as **"value today = best (immediate reward + discounted expected value tomorrow)"**. The maximisation is over the available actions a; the sum is the expected next-state value under the transition kernel.

This is a one-step **accounting identity**, not yet an algorithm: it relates each state's value to its neighbours' values. It says nothing about **how** to solve the resulting system (that is value iteration, policy iteration, Q-learning — deferred to S6); it only states the consistency condition any optimal value must obey.

> **Worked example — one step of the machine.** Suppose working yields reward +1 per day and being broken yields 0, with discount $\gamma = 0.9$. Once we have (by whatever method) the values $V(\mathrm{W})$ and $V(\mathrm{B})$, the Bellman equation for the broken state weighs the two actions:

$$V(\mathrm{B}) = \max\begin{cases} \underbrace{-1}_{\text{repair cost}} + 0.9[0.6\,V(\mathrm{W}) + 0.4\,V(\mathrm{B})] & \text{(repair)} \\ \underbrace{-5}_{\text{replace cost}} + 0.9\,V(\mathrm{W}) & \text{(replace)} \end{cases}.$$

Whichever bracket is larger names the optimal action in the broken state. The equation has done its one job: expressed **value today** as **best reward today plus discounted value tomorrow**.

Common pitfall The discount factor must satisfy $\gamma < 1$ for the infinite-horizon value to be finite — it is exactly the ratio of the geometric series $\sum \gamma^t = 1/(1-\gamma)$ that converges only for $|\gamma| < 1$ (MAT11-1). And the Bellman equation presumes the **Markov property**: if tomorrow depended on the full history, no single number $V(s)$ per state could summarise the future, and the whole framework would collapse.

Exercises

28. With the machine above, $\gamma = 0.9$, rewards $+1$ for W and 0 for B, and the guessed values $V(\mathrm{W}) = 8$, $V(\mathrm{B}) = 4$: evaluate both branches of the Bellman equation for $V(\mathrm{B})$ and decide whether to repair or replace.

29. Explain why a discount $\gamma = 1$ can make the value of a never-ending process infinite, and why $\gamma < 1$ avoids this.

30. Give one real situation where the Markov property is a reasonable modelling assumption and one where it clearly fails, justifying each.